Alice Shang Xu

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EDUCATION

University of California, Santa Barbara *Santa Barbara, CA*

B.S. in Statistics & Data Science, B.A. in Economics | GPA: 3.84/4.0 | Dean's Honors *Sep 2022 – June 2026*

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Regression Analysis, Probability Theory

TECHNICAL SKILLS

Languages: Java, Python, SQL, R, Go
Frameworks & Web: Spring Boot, React, Spring Security, React Router, REST APIs, JWT, Ant Design
Databases & Cloud: PostgreSQL, Spring Data JDBC, AWS (RDS, ECR, App Runner), Google Cloud, Elasticsearch
Tools & Libraries: Git, Docker, pandas, NumPy, scikit-learn, Jupyter, Quarto

EXPERIENCE

Software Automation Intern, Risk Analytics | Orient Securities Company Limited | Hong Kong *Jun – Aug 2024*

- Built **Python automation scripts** to scrape, clean, and refresh equity data for **200+ firms**, reducing update cycles by **40%** and improving monitoring efficiency.
- Implemented **forecasting and valuation workflows** in **VBA** and Excel using reusable functions and structured inputs, supporting ROI analysis and risk scenario planning for Pop Mart (9992.HK).
- Standardized IPO risk assessment datasets and generated structured risk outputs to support repeatable investment screening and research workflows.

Data Analyst / Developer | UCSB Daily Nexus | Santa Barbara, CA *Oct 2024 – Present*

- Led housing affordability analysis using **10 years of UCSB data**, building reusable **R-based analysis scripts** to process, model, and visualize long-term pricing trends.
- Developed reporting workflows for public-facing data stories reaching **20,000+ students**.
- Integrated disparate datasets from university stakeholders and organized **5+ interviews** to improve data quality, reporting context, and technical accuracy.

PROJECTS

OnlineOrder: Spring Boot Food Ordering Web Application

- Built **8+** CRUD REST APIs with **Spring Boot** to support user registration, menu search, ordering, and checkout workflows across core customer interactions.
- Integrated **Spring Data JDBC** and **PostgreSQL on AWS RDS**, ensuring the system handles **300+ concurrent database transactions** with zero connection timeouts.
- Developed a responsive frontend with **React and Ant Design**, implementing **Debounce-based search** and memorized components that reduced redundant re-renders and cut client-side CPU usage by **~25%**
- Containerized the application and deployed it with **AWS ECR** and **AWS App Runner** to improve cloud accessibility.

SocialAI: AI-Based Social Network

- Built social networking application using **React** frontend and **Go** microservices backend deployed on **Google App Engine**, integrating **DALL·E 3 API** for AI-generated image content.
- Engineered **Elasticsearch** cluster on **Google Compute Engine** implementing real-time full-text search across **20,000+ posts** with custom analyzers and aggregations, achieving **sub-second** query response times.
- Designed **RESTful API** architecture handling post creation, user interactions, and feed generation with **JWT-based** authentication and refresh token rotation, serving **400+** concurrent users with **p95** latency under **200ms**.
- Implemented distributed **caching** strategy using **Redis** for frequently accessed posts and user profiles, reducing database load by **60%** and improving feed rendering performance from **1.2s to 400ms**.

Clinical Risk Prediction Pipeline

- Built an end-to-end **machine learning pipeline** in **R** for heart failure risk prediction, including data cleaning, feature engineering, model training, and evaluation.
- Achieved **90% prediction accuracy** with KNN and automated evaluation steps to improve reproducibility and streamline iteration.